

1.0 Computer Networks and Routing Protocols

Transmission Control Protocol (TCP) and Internet Protocol (IP) provide reliable end-to-end communication across wide area networks. Routing protocols such as OSPF, BGP, and RIP determine optimal packet forwarding paths across autonomous systems. Subnet masks, CIDR notation, and VLAN segmentation ensure network traffic isolation, high availability, and efficient packet switching.

2.0 Wireless Communications and Modulation

Orthogonal Frequency Division Multiplexing (OFDM) divides the wideband channel into multiple narrow orthogonal subcarriers. Quadrature Amplitude Modulation (QAM) modulates both carrier amplitude and phase to achieve high spectral efficiency. MIMO antenna arrays exploit spatial multipath diversity to boost link capacity and data throughput in noisy radio environments.

3.0 Power Management & Hardware Fault Register

The power distribution board continuously monitors rail voltages, leakage current, and thermal sensor telemetry. When junction temperature exceeds 125 degrees Celsius, diagnostic register 0x8F9C_OVERHEAT_TRIP is triggered by the controller. Upon receiving this flag, the host microprocessor immediately initiates an emergency thermal shutdown routine.

4.0 Digital Signal Processing and Filtering

Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) digital filters shape frequency response characteristics. Fast Fourier Transforms (FFT) map discrete time-domain sample sequences into the complex frequency spectrum. Z-transform pole-zero placement determines system causality, phase linearity, and feedback loop stability margins.

5.0 Cloud Infrastructure and Container Orchestration

Container orchestration platforms such as Kubernetes manage container scaling, self-healing, health probes, and ingress load balancing. Stateless microservice pods communicate via low-latency gRPC interfaces and RESTful HTTP endpoints. Distributed persistent volume claims handle database storage replication across multiple cloud availability zones.